

Software Testing and Maintenance

Spring, 2026



INTRODUCTION

- Software Testing is a **critical** element of developing quality software systems
- It is a systematic approach to **judge quality and discover bugs**
- This course presents the **theory and practice** of software testing and Maintenance

INTRODUCTION

- Topics covered include:
 - **Black-box** and **white-box** testing, and related **test case** generation
 - Testing team and testing documentation
 - Tools for software testing
 - Performance testing basics
 - Testing in the Software Process

COURSE OBJECTIVES

- Understand the **concepts and theory** related to software testing and quality assurance
- Understand the relationship between **black-box and white-box** testing, and know how to apply as appropriate
- Understand different **testing techniques and process** used for developing test cases and evaluating test adequacy
- Learn to use **automated testing tools** efficiently and effectively
- Understand current state of software testing and maintenance

GRADING

- Daily performance (40%)
- Final Exam (60%)

Academic Honesty in a Digital World



Penalties for Cheating

If you cheat in this class, you will fail the class.

TEXTBOOK

➤ Software Testing Principles and Practice

Second Edition, English language

*S. Brown, J. Timoney, T. Lysaght and D. Ye
China Machine Press, 2019*

- The book is developed over ten years and reflects the experience in **industry and lecturing** from authors;
- The book is the results of extracting and interpreting test techniques from **a wide classic books and standards**.
- By providing extensive **worked examples**, the book aims to provide a solid basis for both understanding and applying various test techniques



CONTENTS

➤ Software Testing Principles and Practice

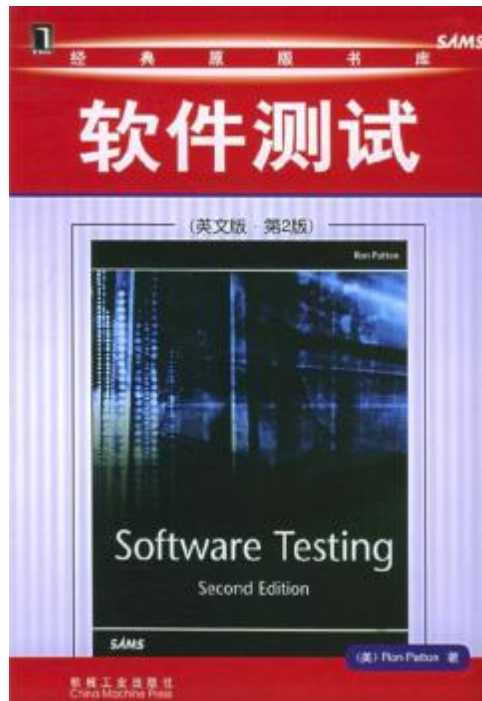
1. Introduction to Software Testing (*Chapter 1*)
2. Testing in the software Process (*Chapter 9*)
3. Black-Box Testing (*Chapter 2&3*)
4. White-Box Testing (*Chapter 2&3*)
5. Test Team & Test Documentation
6. Unit Testing (*Chapter 3,4&5*)
7. Integration Testing (*Chapter 6*)
8. System Testing (*Chapter 7*)
9. Software Test Automation (*Chapter 8*)
10. Software Maintenance



REFERENCES

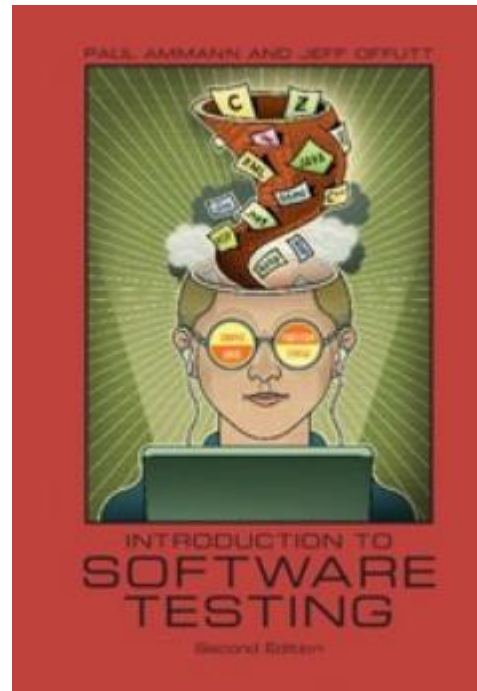
Software Testing

Second Edition, Ron Patton
China Machine Press, 2006



Introduction to Software Testing

Second Edition, Paul Ammann & Jeff Offutt
Cambridge University Press, 2018



How Google Test Software

James Whittaker, Jason Arbon, Jeff Carollo
Posts and Telecommunications Press, 2016



<http://www.cs.gmu.edu/~offutt/softwaretest/>

References Book

软件测试方法和技术
(第4版)

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